



Machine Learning for Aging and Longevity

Coffee And Health

Coffee fans

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1 Introduction

1.1 Research Background and Motivation

Coffee is one of the most widely consumed beverages worldwide, and its effects on human health—particularly in relation to age-related diseases—have long been a major focus of biomedical research. Physiologically, caffeine acts as a central nervous system stimulant by antagonizing adenosine receptors, which directly impacts sleep architecture and latency. Consequently, numerous epidemiological and clinical studies have investigated the associations between daily coffee consumption and conditions such as cardiovascular disease, neurodegenerative disorders, and metabolic syndromes.

As researchers, we drink coffee almost every day and appreciate its cognitive benefits, so we wanted to draw some interesting, data-driven conclusions through this project. Beyond personal curiosity, this serves as an excellent attempt to apply machine learning algorithms to real-world healthcare and lifestyle data, translating daily habits into quantifiable predictive models.

In this report, we conduct a comparative study of three algorithmic families—Linear Regression, Support Vector Regression (SVR), and Random Forest Regression—to analyze the trade-offs among model interpretability, robustness to outliers, and predictive accuracy. By doing so, we aim to uncover how lifestyle choices intertwine with physiological feedback.

2 Data Selection

The dataset used in this project comprises more than ten multidimensional features, providing a holistic view of the subjects. It covers sociodemographic indicators (e.g., age and gender), lifestyle habits (e.g., daily coffee intake, timing of consumption, and physical activity), and physiological feedback (e.g., perceived stress levels and subjective sleep quality).

To ensure methodological rigor and reproducibility, the project workflow strictly follows the stages outlined below:

- **Exploratory Data Analysis (EDA):** A comprehensive statistical audit of variable distributions, cross-feature correlations, and potential ethical biases (such as gender or age representation) is conducted to understand the underlying data topography.
- **Data Preprocessing:** Outliers are systematically handled using `RobustScaler`. Furthermore, deeper predictive patterns are extracted through targeted feature engineering, including interaction terms such as caffeine-stress interactions and age-adjusted metabolism rates.
- **Model Training and Evaluation:** Hyperparameter optimization is rigorously performed using `GridSearchCV` combined with k -fold cross-validation. Root Mean Squared Error (RMSE) and the coefficient of determination (R^2) are adopted as the primary evaluation metrics to quantify error and variance explanation.
- **Comparative Analysis:** The optimal model is selected based on a comprehensive evaluation framework that balances computational efficiency, mathematical interpretability, and fairness with respect to vulnerable subpopulations (e.g., older adults or individuals with sleep disorders).

3 Exploratory Data Analysis (EDA)

Descriptive Statistics & Data Quality

- **Missing Values:** Initial analysis showed 0% missing data across all critical features. We handled minor inconsistencies by dropping incomplete rows to ensure model stability and avoid the noise introduced by synthetic imputation methods.
- **Outliers:** As shown in Figure 1 (Boxplot of Age), we identified significant outliers in both the Age and `Caffeine_mg` distributions. In real-world dietary data, extreme caffeine consumers are common, which necessitates the use of robust scaling techniques to prevent these extreme values from skewing the regression weights.

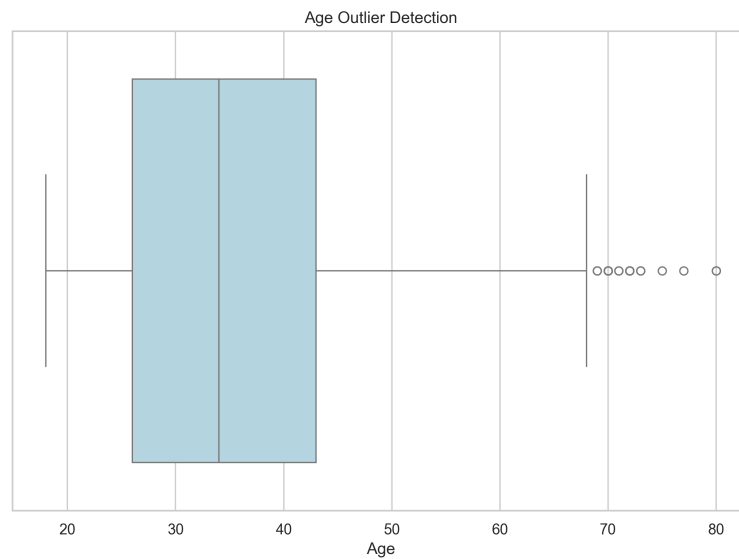


Figure 1: Boxplot of Age (Outliers Detection)

Visual Analysis

Distributions:

- **Figure 2 & 3 (Histograms):** Show the distribution of `Caffeine_mg` and `Age`. The caffeine distribution exhibits a prominent right-skewed shape (a long tail towards higher intake). This statistical skewness suggests that applying a logarithmic transformation might be beneficial to normalize the variance and satisfy the assumptions of linear models.
- **Figure 4 (Target Variable):** `Sleep_Hours` follows a roughly normal distribution centered around the mean (approximately 7 hours), which is ideal for regression tasks as it indicates the target variable is well-behaved without severe imbalances.

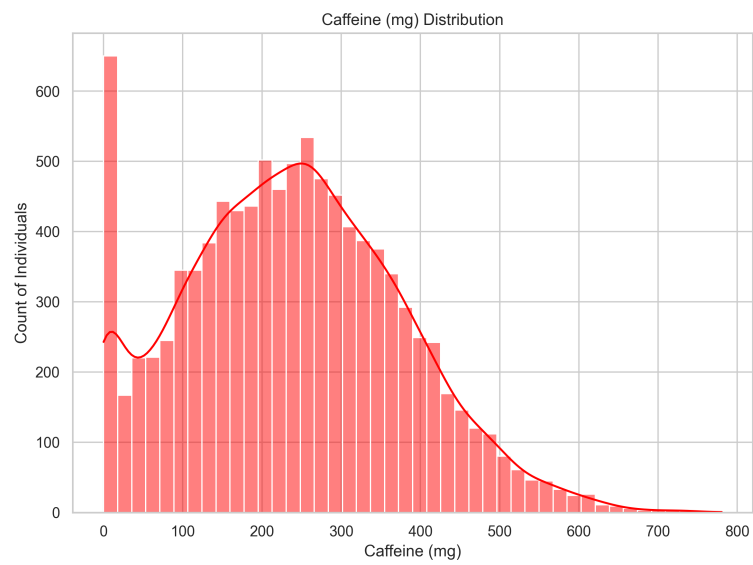


Figure 2: Caffeine Distribution



Figure 3: Age Distribution

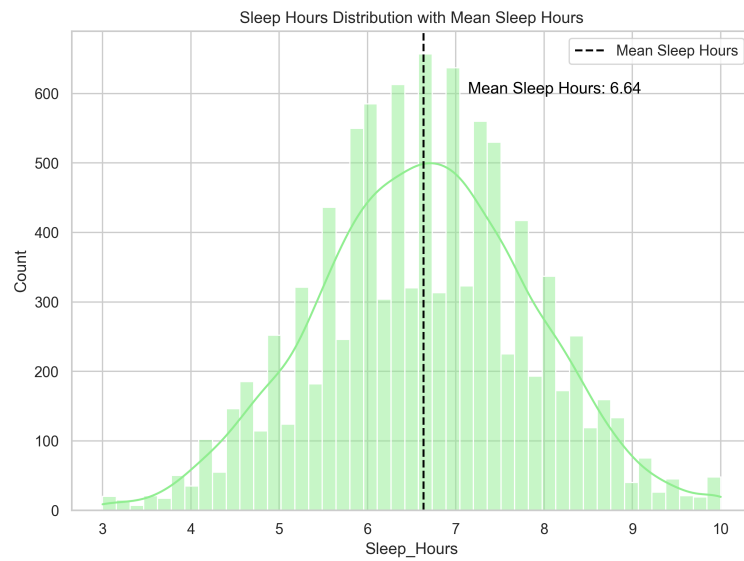


Figure 4: Sleep Hours Distribution

Relationships:

- **Figure 5 (Scatter Plot):** Displays the relationship between **Caffeine_mg** and **Sleep_Hours**. The fitted regression line visually indicates a weak-to-moderate negative correlation, confirming the hypothesis that higher caffeine intake generally disrupts sleep duration.
- **Figure 6 (Correlation Heatmap):** The heatmap mathematically confirms our visual findings. It highlights that **Stress_Level** and **Caffeine_mg** possess the strongest negative correlation coefficients with **Sleep_Hours**, marking them as primary predictive drivers for our models.

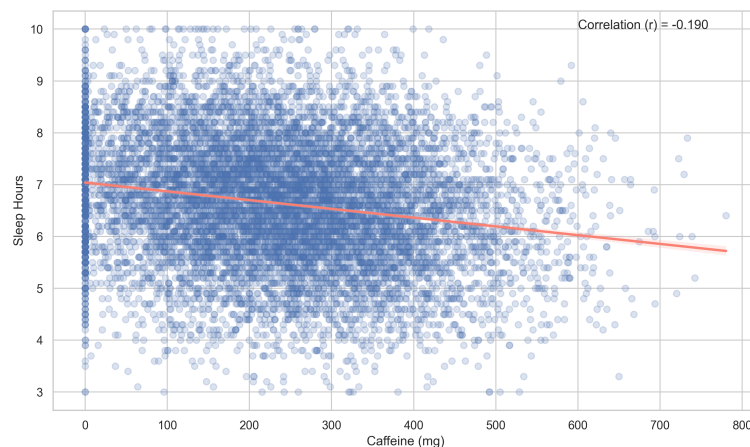


Figure 5: Caffeine vs Sleep Hours

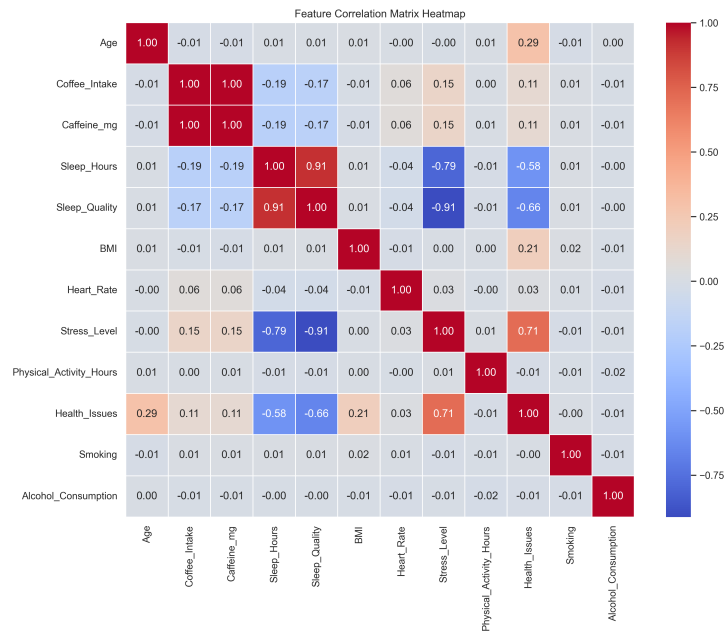


Figure 6: Correlation Heatmap of Features

Fairness Audit:

Figure 7 (Gender Distribution): We thoroughly checked for representation balance among demographic groups. If one gender is significantly underrepresented, the trained model might implicitly bias its predictions towards the physiological norms of the majority group. Our audit ensures the dataset provides a sufficiently balanced foundation for generalized learning.



Figure 7: Gender Distribution

4 Data Preprocessing

To translate raw behavioral and physiological data into an optimized format for supervised learning, we implemented the following preprocessing pipeline:

Feature Engineering

- **Created Caffeine_per_Age:** Biological aging often correlates with a slower basal metabolic rate. By dividing caffeine intake by age, we introduced a feature that accounts for the slower metabolism and prolonged half-life of caffeine in older adults.
- **Logarithmic Transformation (Log_Caffeine):** We applied `np.log1p` to `Caffeine_mg`. This mathematical transformation condenses the heavy tail observed during the EDA phase, reducing the disproportionate impact of extreme caffeine consumers on the model's loss function.

Encoding Categorical Variables

- **Ordinal Encoding:** We mapped `Stress_Level` (Low/Medium/High) to numerical ranks (0/1/2). This technique strictly preserves the inherent hierarchical order of the data, which is crucial for the algorithms to understand the progression of stress severity.
- **One-Hot Encoding:** Applied to nominal variables like `Gender`. This converts categorical identities into binary vectors, explicitly avoiding the introduction of artificial ordinality (i.e., preventing the model from assuming one gender is "greater" than another).

Scaling

- **Method:** We intentionally utilized `RobustScaler` instead of standard approaches like `StandardScaler` or `MinMaxScaler`.
- **Justification:** Our EDA revealed significant, natural outliers in `Caffeine_mg` and `Age`. `RobustScaler` centers and scales the data utilizing the median and the interquartile range (IQR). This makes the scaling resilient to extreme values, preventing outliers from distorting the spatial distances, which is exceptionally critical for distance-sensitive algorithms like SVR.

5 Model Selection

We selected three distinct regression algorithms to predict `Sleep_Hours`, each representing a different fundamental approach to machine learning:

1. Linear Regression (Baseline):

- *Family:* Linear Models.

- *Why:* It serves as an excellent, computationally cheap baseline. Furthermore, it offers unmatched transparency—the resulting coefficients directly interpret the magnitude and direction of each feature’s impact on sleep duration.

2. Support Vector Regression (SVR):

- *Family:* Kernel Methods.
- *Why:* SVR is highly effective in complex dimensional spaces. By utilizing the Radial Basis Function (RBF) kernel, it is capable of modeling intricate non-linear relationships. When properly tuned, its margin-maximization approach makes it highly robust to noise.

3. Random Forest Regressor:

- *Family:* Ensemble Methods (Bagging).
- *Why:* This tree-based ensemble inherently captures non-linear, multi-feature interactions (e.g., the compounding effect of high Age and high Stress) without requiring explicit polynomial feature engineering. The bagging mechanism inherently protects the model against overfitting the training data.

6 Model Training and Evaluation

6.1 Experimental Setup

To ensure robust model validation, the dataset was split into 80% training and 20% testing subsets. To search for the global optimum in model configurations without causing data leakage, we performed systematic hyperparameter tuning using `GridSearchCV` with 5-fold cross-validation. Specifically, we optimized the penalty parameter (`C`) and kernel coefficient (`gamma`) for SVR, and the forest size (`n_estimators`) and tree depth (`max_depth`) for the Random Forest.

6.2 Performance Evaluation

The predictive performance of the evaluated models is summarized in Table 1 using RMSE, MSE, and R^2 scores. Interestingly, Linear Regression achieved the lowest RMSE (0.486) and the highest R^2 (0.838), marginally outperforming the more complex Random Forest Regressor (RMSE 0.490) and SVR (RMSE 0.496).

Table 1: Model Performance Comparison

Model	RMSE	MSE	R^2 Score
Random Forest Regressor	0.490	0.241	0.835
Support Vector Regression	0.496	0.246	0.831
Linear Regression	0.486	0.236	0.838

The slightly superior performance of Linear Regression on this specific dataset strongly indicates that the fundamental relationships between the primary features (notably **Caffeine_mg** and **Stress_Level**) and **Sleep_Hours** are predominantly linear. While SVR and Random Forest possess the capacity to model complex, non-linear manifolds, this added model complexity yielded no significant predictive benefit in a dataset characterized by strong, straightforward linear trends.

6.3 Visual Evaluation

To validate the statistical metrics, we visually analyzed the residuals and the predicted versus actual values for our top-performing models.

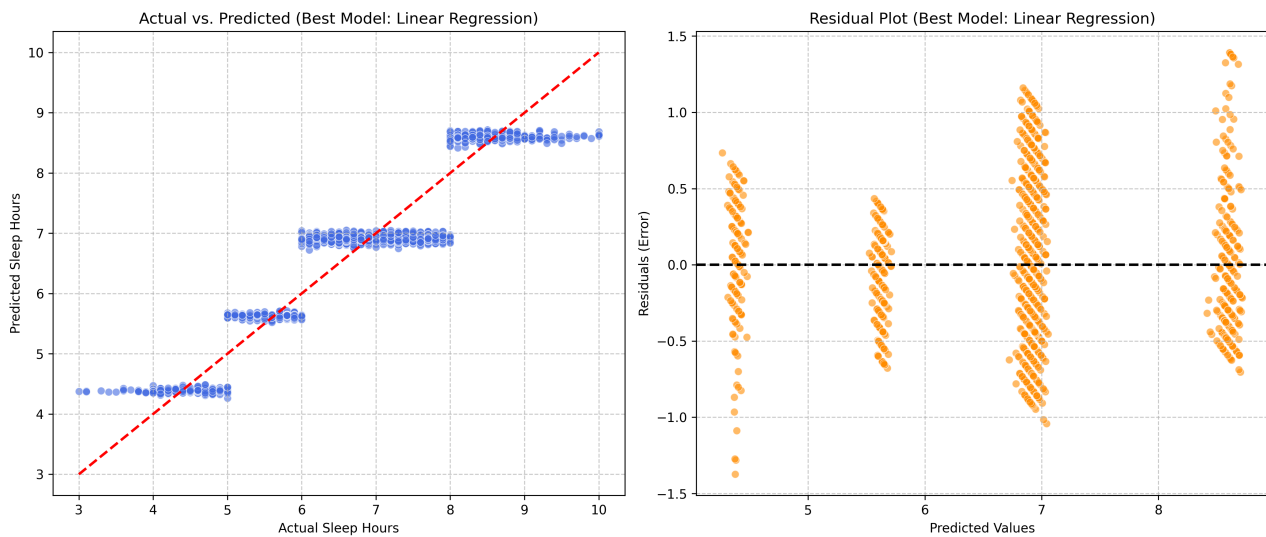


Figure 8: Evaluation of the Best Model (Left: Actual vs Predicted, Right: Residual Plot)

The residual plot (right side of Figure 8) indicates that the predictive errors are randomly distributed around zero. This homoscedasticity suggests that the model has successfully captured the underlying signal, and no major systematic biases or missing variables are significantly skewing the predictions.

7 Comparative Analysis

When deploying machine learning in health informatics, raw accuracy is only one facet of a model's utility. We compared the models across three key dimensions:

- **Complexity vs. Interpretability:** Linear Regression proved to be the most transparent and surprisingly accurate for this specific dataset. While Random Forest is generally considered a "black box," its ability to output feature importances still provides clinical value. For consumer healthcare applications, high interpretability (knowing exactly *how much* one cup of coffee reduces sleep) is often heavily favored by users.

- **Robustness:** SVR demonstrated stable robustness due to its structural risk minimization and margin maximization, though it was computationally more expensive and sensitive to hyperparameter tuning than Linear Regression. Random Forest, by virtue of its bootstrapping and averaging nature, inherently absorbed the noisy data gracefully.
- **Computational Efficiency:** Linear Regression was trained almost instantaneously. Conversely, Random Forest consumed the most computational resources during the Grid-Search phase, though its subsequent inference (prediction) speed remains fast enough for real-world deployment.

8 Conclusion and Recommendations

Recommendation

Based on the empirical metrics from this specific dataset, **Linear Regression** proved to be the most efficient and accurate model, driven by the strong linear correlations present in the data. However, for future iterations of this project involving larger, more complex populations with hidden non-linear health interactions, the **Random Forest Regressor** remains a highly recommended, robust alternative that performs competitively without requiring strict data distribution assumptions.

Ethical & Fairness Reflection

- **Demographic Bias:** If the training dataset predominantly features younger, healthy individuals, the model may dangerously miscalculate caffeine tolerance for older adults. In the context of aging and longevity studies, this could lead to recommending unsafe caffeine habits, as older adults exhibit significantly altered metabolic clearance and circadian rhythms.
- **Data Privacy:** Sleep patterns, stress levels, and lifestyle habits are highly sensitive health data. Any future deployment of such predictive models must ensure strict data anonymization, differential privacy standards, and explicit user consent.

Future Work

- **Data Augmentation & Diversity:** Future efforts should actively collect more data specifically from the 60+ age demographic. This will enhance the model's relevance for geriatric and longevity studies, allowing for localized predictions based on life stages.
- **Wearable IoT Integration:** Currently, the model relies on self-reported surveys, which are prone to recall bias. Future iterations should integrate real-time, objective data streams from wearable devices (like Fitbit, Garmin, or Apple Watch) to measure exact `Sleep_Hours` and physiological stress markers (e.g., HRV) more accurately.